Numerical Integration

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Lesson 4

5. Simpson's Rule

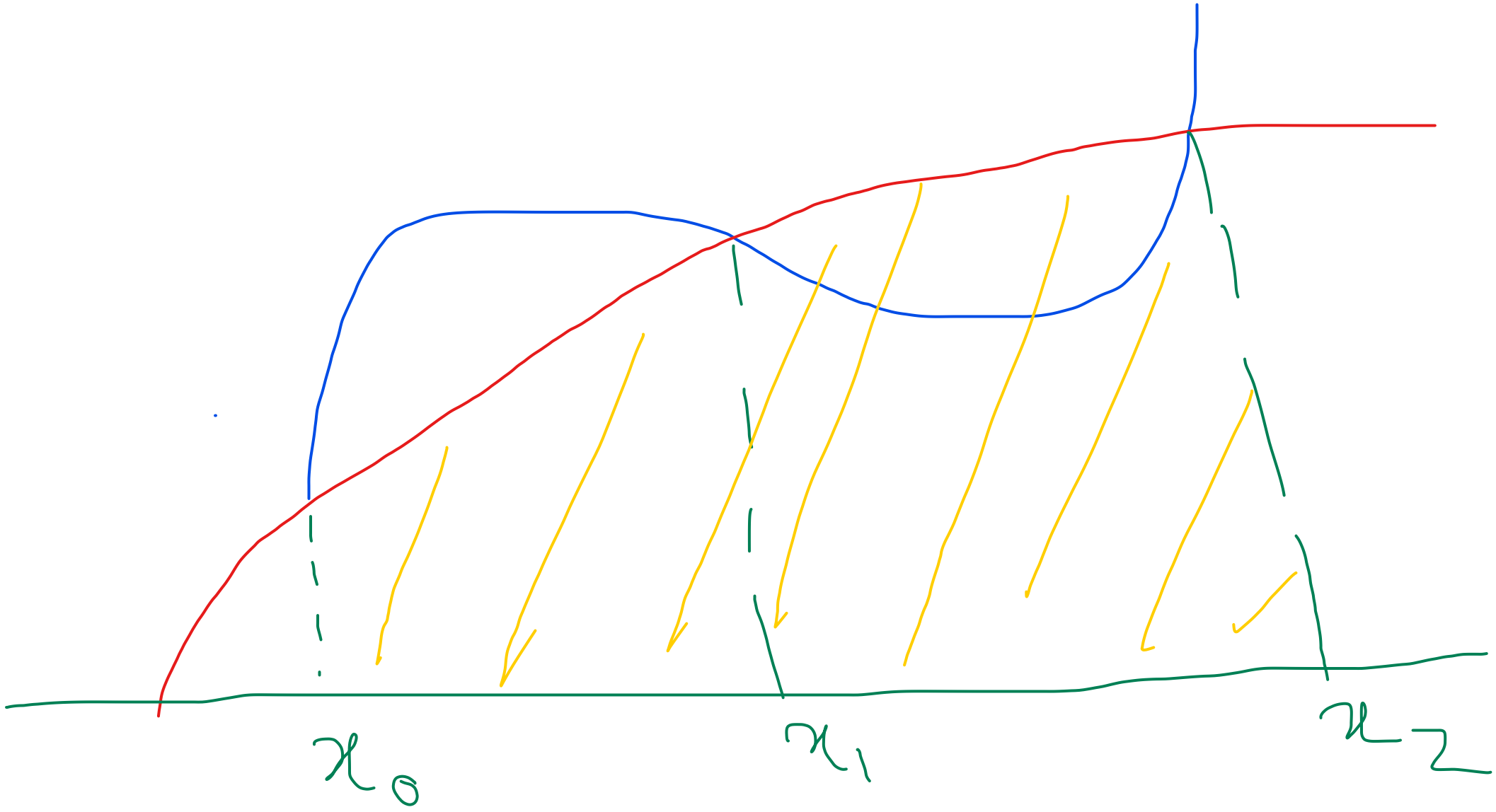
1. Introduction

- The **Simpson's Rule** is a numerical integration method used to approximate the value of a definite integral.
- Suppose we want to evaluate

$$I = \int_a^b f(x)\,dx$$

when finding the exact integral is difficult or when only a set of discrete function values is available.

- The basic idea of Simpson's Rule is to approximate the function using a **quadratic polynomial (parabola)** rather than a straight line.


x_0
x_1
x_2

5.1. Simpson's 1/3 Rule

- The Trapezoidal Rule approximates a curve using a straight line.
- Simpson's 1/3 Rule approximates the curve using a **quadratic polynomial**.

Consider three equally spaced points:

$$x_0, \qquad x_1, \qquad x_2.$$

The spacing between consecutive points is

$$x_1 - x_0 = h$$

and

$$x_2 - x_1 = h.$$

Therefore,

$$x_2 - x_0 = 2h.$$

Let

$$f_0 = f(x_0), \qquad f_1 = f(x_1), \qquad f_2 = f(x_2).$$

Simpson's $1/3$ Rule is

$$\int_{x_0}^{x_2} f(x)\,dx \approx \frac{h}{3}\left[f_0 + 4f_1 + f_2\right]$$

or

$$\int_{x_0}^{x_2} f(x)\,dx \approx \frac{h}{3}\left[f(x_0) + 4f(x_1) + f(x_2)\right]$$

The coefficient pattern is

$$1,\ 4,\ 1$$

Error formula

$$\frac{h^5}{50} f^4(\xi)$$

Question

Use Simpson's $1/3$ Rule to approximate

$$\int_0^2 (x^2 + 1)\, dx.$$

- Identify the points

Here, a=0,b=2

Using three points,

$$x_0 = 0, \qquad x_1 = 1, \qquad x_2 = 2.$$

Therefore,

$$h = 1.$$

Step 2: Calculate the function values

Given

$$f(x) = x^2 + 1.$$

Therefore,

$$f(0) = 0^2 + 1 = 1$$

$$f(1) = 1^2 + 1 = 2$$

$$f(2) = 2^2 + 1 = 5.$$

Step 3: Apply Simpson's $1/3$ Rule

$$I \approx \frac{h}{3}[f(x_0) + 4f(x_1) + f(x_2)]$$

Substituting the values,

$$I \approx \frac{1}{3}[1 + 4(2) + 5]$$

$$= \frac{1}{3}[1 + 8 + 5]$$

$$= \frac{14}{3}.$$

Therefore,

$$\boxed{I \approx 4.6667}$$

The exact value is also

$$\int_0^2 (x^2 + 1)dx = \frac{14}{3} = 4.6667.$$

So, in this example, Simpson’s Rule gives the exact answer because the function is a quadratic polynomial.

5.2. Composite Simpson's 1/3 Rule

- The basic Simpson's Rule uses only **two subintervals**.
- For larger intervals, we divide [a,b] into several equally spaced subintervals and apply Simpson's Rule repeatedly.
- This is called the **Composite Simpson's 1/3 Rule**.
- Suppose the interval is divided into n equal subintervals.

Then,

$$h = \frac{b-a}{n}.$$

For Simpson's $1/3$ Rule,

$$\boxed{n \text{ must be even}}$$

because each application of Simpson's Rule uses two subintervals.

The formula is

$$\int_a^b f(x)\,dx \approx \frac{h}{3}\left[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\right]$$

Pattern of Coefficients

The coefficients in Composite Simpson's 1/3 Rule follow the pattern

1,4,2,4,2,4,…,2,4,1

Therefore:

- First value → coefficient 1
- Interior odd-indexed values → coefficient 4
- Interior even-indexed values → coefficient 2
- Last value → coefficient 1

For example, if n=6:

1,4,2,4,2,4,1

Error formula

$$E_s = \frac{h^4(b-a)}{180} f^4(\xi)$$

Need to take max $f^4(\xi)$

Error bound for Simpson's rule

$$E_s \leq \frac{h^4(b-a)}{180} f^4(\xi)$$

Question

Use the Composite Simpson's $1/3$ Rule to approximate

$$\int_0^6 (x^2 + 1)\, dx$$

using $n = 6$ subintervals.

Step 1: Calculate the step size

$$h = \frac{b-a}{n}$$

$$h = \frac{6-0}{6} = 1.$$

Step 2: Construct the table

Given

$$f(x) = x^2 + 1.$$

i	x_i	$f(x_i)$	Coefficient
0	0	1	1
1	1	2	4
2	2	5	2
3	3	10	4
4	4	17	2
5	5	26	4
6	6	37	1

Step 3: Apply the formula

$$I \approx \frac{h}{3}[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + 4f(x_5) + f(x_6)]$$

Substituting,

$$I \approx \frac{1}{3}[1 + 4(2) + 2(5) + 4(10) + 2(17) + 4(26) + 37]$$

$$= \frac{1}{3}[1 + 8 + 10 + 40 + 34 + 104 + 37]$$

$$= \frac{234}{3}$$

$$\boxed{I = 78}$$

Thus,

$$\boxed{\int_0^6 (x^2 + 1)\, dx \approx 78}$$

Again, the result is exact because $f(x) = x^2 + 1$ is a quadratic function.

Find the error value

$$\int_1^5 \frac{1}{x} dx \quad for\ n = 4$$

$$f(x) = \frac{1}{x}$$
$$f'(x) = -x^{-2}$$
$$f''(x) = 2x^{-3}$$
$$f'''(x) = -6x^{-4}$$
$$f^4(x) = \frac{24}{x^5}$$

$$max|f^4(x)| = \frac{24}{1^5} = 24$$

The largest vale that $\frac{24}{x^5}$ will take as $x$ranges from 1 to 5

$$E_s \leq \frac{h^4(5-1)}{180} \times 24$$

$$h = \frac{b-a}{n} = \frac{5-1}{4}$$

$$E_s \leq \frac{(\frac{4}{4})^4(5-1)}{180} \times 24$$

$$E_s \leq \frac{4}{180} \times 24 \approx 0.53$$

Simpson's 1/3 Rule vs Trapezoidal Rule

Feature	Trapezoidal Rule	Simpson's (1/3) Rule
Approximation	Straight lines	Parabolas
Number of points	2 for basic rule	3 for basic rule
Basic formula	$\frac{h}{2}[f(x_0)+f(x_1)]$	$\frac{h}{3}[f(x_0)+4f(x_1)+f(x_2)]$
Polynomial used	Linear	Quadratic
Subinterval requirement	No special requirement	(n) must be even
Accuracy	Generally lower	Generally higher
Coefficients	(1,1)	(1,4,1)

Examples

Use Simpson's 1/3 rule to approximate

$$\int_0^1 \frac{1}{1+x^2}\,dx$$

Use Composite Simpson's 1/3 rule to approximate

$$\int_0^2 e^{-x^2}\,dx$$

For $n = 4$